RUDRA KARADKHELKAR

rudrak@mit.edu • linkedin.com/in/rudra-karadkhelkar • Cambridge, MA

EDUCATION

Massachusetts Institute of Technology Cambridge, MA
B.S. in Computer Science and Artificial Intelligence *Expected June 2030*

- **Relevant Coursework:** 6.100A (Introduction to Computer Science and Programming in Python), Multivariable Calculus & Linear Algebra (18.02), Classical Mechanics (8.01), Solid-State Chemistry (3.091)

RESEARCH & EXPERIENCE

MIT UROP, Cybersecurity at MIT Sloan (CAMS) Cambridge, MA
Undergraduate Researcher: Advisors - Dr. Ranjan Pal, Dr. Michael Siegel *Aug 2026 – Present*

- Deriving two-dimensional graph centrality metrics that jointly weight node and edge importance to identify critical assets in infrastructure networks (power grids, transportation), a formulation not yet in the literature
  - Validating the metrics on real-world network topologies supplied by a CAMS partner organization

CGI Federal Lafayette, LA
Technical/Project Lead & AI Intern *Jun 2026 – Aug 2026*

- Led the redesign of CGI's shipment-tracking platform, consolidating 3 systems into 1 nationwide data hub
- Built the platform's database design, 5+ screens, and automated scheduling logic end to end, eliminating manual coordination on shipment requests
- Built an AI agent that auto-drafts requests from Microsoft Teams Q&A meetings; ran weekly Agile sprints

Data Analysis/Software Intern *Jun 2025 – Oct 2025*

- Centralized tracking of 10-15 licenses (Azure, GitHub, Copilot, etc.) across 8 business units into one executive dashboard, replacing hours of manual reconciliation; earned a 2-month extension to expand the project's scope

PROJECTS

CyberDefender: Autonomous AI Network Defense Agent *In Progress*
Self-proposed; approved and supervised by UROP advisor Dr. Ranjan Pal (MIT CAMS); MVP targeted December 2026

- Building a multi-agent SOC on an isolated Linux home lab: three read-only agents monitor logs, network traffic, and file activity, and an overseer agent with admin rights classifies threats and contains them
- Self-healing loop tests candidate fixes on an isolated clone VM before deploying to the live system; model-agnostic harness (swappable frontier/local LLMs) with a typed-decision layer validating every action
- Red-teaming from a Kali Linux attacker VM and building a benchmark for detection rate, false positive rate, fix success rate, and system degradation

Jarvis: Self-Hosted AI Automation System

- Self-hosted a Linux server running 4+ specialized AI subagents that autonomously handle daily productivity tasks
- Hardened with full-disk encryption, zero exposed remote ports, secret scanning, and hard safety gates on every agent

LEADERSHIP & ACTIVITIES

MIT Satellite Team Cambridge, MA
Software Developer *Sep 2026 – Present*

- Learning NASA JPL's F' C++ flight software framework, which the team is adopting for its next satellite launch

MIT AI Alignment (MAIA): AI Safety Fundamentals (AISF) Reading Group *Sep 2026 – Present*

FIRST Robotics, Team 3616 "Phenomena" Lafayette, LA
Lead Programmer, President, Project Manager *May 2021 – Jul 2026*

- Lead Programmer for 2 seasons: C++/Java subsystem control (swerve drive, elevator, arm, PID tuning), then autonomous routines and vision tracking
- **2025 FIRST Dean's List Award Winner**: 1 of 10 selected from 95,000+ students worldwide; FIRST podcast guest
- Elected President for 2 terms, then Project Manager (highest role), directing 5 subteams and 20+ members

CAPE Satellite Program, University of Louisiana at Lafayette Lafayette, LA
Backend Developer *Mar 2024 – Mar 2026*

- Only high school student on the CubeSat team, working with NASA engineers to develop and launch picosatellites into orbit
- Self-taught Django to build ground-station communication endpoints and designed PostgreSQL databases for satellite telemetry; authored the team's standard onboarding plan

SKILLS

Languages: Python, C++, Java, SQL, PowerShell
Security & Systems: Linux, Kali Linux, VirtualBox, Tailscale, Git
Backend: Django, Django Ninja, PostgreSQL, SQLite
AI & Automation: LLM agents, Local LLMs, Claude Code, n8n, Copilot Studio, Power Automate, Power Apps, Power BI, SharePoint